This DCC-Dapp is a long-running program because the completion of the required courses may take as long as four years. More important, the university may want a historical ledger of DCC timeline and engagement. Users may use this DCC-Dapp as their progress sheet through the certificate program. With this context, here are the numbered functions, as shown in figure 11.3:

1. *Register.* A user identified by a 256-bit decentralized account address registers their person number associated with the university or institution. This operation is done only once, at the beginning of the process.
2. *Log in.* For subsequent access to the DCC-Dapp, login with the person number and a matching decentralized identity is required; otherwise, the login will be rejected.
3. *Add a course.* To add a course to be considered for the certificate program, the user enters the category, course, and grade in the course.
4. *Check eligibility.* This function determines the user's standing in the certificate program in five criteria: four course categories and overall GPA.
5. *Replace a course.* This function replaces a course in the certificate program. The user probably will add a different course to improve their GPA or update their grade after retaking the course.

These operations will guide the design of the smart contract for the DCC-Dapp. Other support functions may be required to complete the design of the smart contracts.

11.4.2 Contract diagram